SULTAN AL NIYADI

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SUMMARY

Corporate Applications Engineer with 3 years of hands-on experience in sensors, analog circuit design, microcontrollers, and power systems. Skilled in hardware and software applications, delivering accurate, high-quality client engagement. Currently at Microchip Technology, supporting customers in a global semiconductor environment.

EDUCATION

B.S.E., Electrical Engineering **Spring 2019 – Fall 2022**
Arizona State University, Tempe, Arizona
Ira. Fulton Schools of Engineering

M.E., Engineering Management **January 2025 – December 2026**
Arizona State University, Tempe, Arizona
Ira. Fulton Schools of Engineering

TECHNICAL SKILLS

Project Management • Power system Analysis • Electrical design • C/C++ • Digital Design • Leadership • Team Management • Time Management • Client Engagement • Financial engineering • Risk Management • Root Cause Analysis • Systems engineering

Hardware and software

Oscilloscope/Logic Analyzer • LTSpice • Cadence Virtuoso • SOLIDWORKS • Autodesk Fusion 360 • PowerWorld simulator • Altium Designer • MPLABX • Microsoft VSCode • JMP • BitBucket • Claude

PROFESSIONAL EXPERIENCE

Corporate Applications Engineer April 2023 – Present
Company: Microchip Technology Inc., Arizona, United States

- Communicate with clients globally to provide technical support for Microchip products.
- Develop and deploy custom AI solutions designed to significantly improve client engagement, elevate technical support, and streamline project management workflows.
- Use Microsoft VScode for code generation and programming
- Utilize GIT/Bitbucket for project tracking and version control.
- Design PCB boards using Altium Designer.
- Simulate analog circuits and validated designs using the MPLAB Mindi Analog Simulator.
- Engage with client projects, providing technical assistance and management oversight.
- Create and publish technical articles for the internal knowledge base.

Students Association Manager August 2021 – December 2022
Company: United Arab Emirates Student Association, Tempe, Arizona, United States

- Led the Emirati student community in Tempe, AZ, serving as the primary representative.
- Initiated, implemented, and completed collaborative projects for the International Office at Arizona State University.
- Managed an annual budget to successfully fund activities and organize community events for all student members.

PROJECTS

Pressure Sensor reading using 4-20mA Loop Driver October 2024 – August 2026

- **Engineered** a 5-stage, low-current demonstration system, processing differential sensor signals through a differential amplifier topology into a microcontroller’s ADC.
- **Developed** state machine firmware to manage multiple operational states, interface with an onboard display for real-time pressure and current readouts, and drive the DAC to control the system's loop current.
- **Integrated** a power monitor to track system voltage and current, enabling an auto-calibration feature to continuously correct any missing or inaccurate values from the MCU.
- **Created** an interactive GUI utilizing Claude AI to visually represent the signal chain, schematic blocks, and technical resources for high-level presentations to clients and engineering teams.
- **Designed** the project's PCB boards using Altium Designer, programmed the microcontroller using Microsoft VScode, and performed hands-on component soldering and debugging.
- **Result:** Delivered a fully functional differential signal to 4-20mA loop driver board.

Large Systems Failure Analysis & Literature Review May 2026 - July 2026

- Conducted a comprehensive literature review analyzing the root causes of failure in complex, large-scale engineering systems.
- Applied enterprise modeling and systems engineering frameworks to evaluate risk management practices.
- Synthesized industry data to propose optimized management and operational strategies that mitigate risk and reduce project failure rates.